

THE CHALLENGE

إنتاج الفيديوهات التعليمية بلغة الإشارة باستخدام الذكاء الاصطناعي

AI-Powered Sign Language Academic Video Generator

The Ministry of Education wants to explore how AI could transform verified academic content into short, engaging educational videos presented in Arabic Sign Language.

SOURCE



UNDERSTAND



STRUCTURE



GENERATE



VALIDATE



SIGN VIDEO

**ARABIC SIGN LANGUAGE**

- Primary target language for the generated output.
- Preserve educational meaning and intent, not a literal word-for-word translation.
- Consider terminology, vocabulary limits and uncertain translations; flag rather than hide unsupported segments.
- Propose and justify your own fallback approach.

**LOCAL AI MODELS ONLY**

- Core inference (understanding, transformation and sign generation) must run locally within the approved environment.
- External generative AI APIs must not perform these core stages.
- Be ready to explain model family, size, hardware, performance and why your approach suits local deployment.

Episode 1: Cells

Secure

Source: Grade 6 Science, Ch. 3 (Verified) · Local model only

SIGN LANGUAGE TRACK



Episode 1: Cells · primary track: Arabic Sign Language



00:14 / 00:40 Glossary: 3 terms

REVIEW FLAG

Term "mitochondria" signed with low confidence, teacher review suggested.

Illustrative concept only · not the UI candidates are expected to reproduce**KEY REQUIREMENT: SOURCE GROUNDING**

The solution must not introduce educational facts that cannot be traced back to the approved source material. Every generated episode should remain traceable to its source.

YOUR CHALLENGE

كيف ستقوم ببناء الحل؟

How Would You Build It?

Design the end-to-end solution and be prepared to explain and defend your technical decisions.

01

Solution Architecture

Present the end-to-end architecture, from source ingestion through to sign-language delivery. Explain the major components, how they interact, and why you selected them.



02

Role of AI

Explain where AI should be used and where deterministic, rule-based or conventional approaches may be more appropriate. Identify where human-in-the-loop checkpoints add value.



03

Model Selection & Local Deployment

Which locally deployable models or model families would you consider, and why? Discuss size, infrastructure requirements, performance and accuracy trade-offs.



04

Content Understanding

Explain how the solution would understand verified academic material, identify important concepts, preserve factual accuracy and structure a coherent episode.



05

Sign-Language Generation

Explain how educational meaning would be transformed into Arabic Sign Language output. Consider vocabulary coverage, terminology, meaning preservation and how uncertain content would be handled.



The brief is intentionally open.

There is no single correct answer. We want to see how you approach the problem, what you choose to build, and how you defend your technical decisions.

ALSO BE PREPARED TO DISCUSS

PERFORMANCE & EVALUATION

SECURITY & DATA HANDLING

SOURCE VERIFICATION & DATA QUALITY

DEPLOYMENT & SCALABILITY

PROTOTYPE CHALLENGE



"Produce at least one complete short educational episode in Arabic Sign Language, fully traceable to its source material."

The prototype does not need to represent a production-ready platform. We are assessing your ability to turn an ambiguous AI problem into a functioning technical solution and to explain the decisions behind it.

PROTOTYPE JOURNEY

01

Source

02

Understand

03

Structure

04

Generate

05

Validate

06

Sign Video

YOUR DELIVERABLES: TWO WEEKS FROM RECEIPT TO SUBMIT

01

Working Prototype

End-to-end functioning demo

02

Max 4 Slides

Problem, architecture, decisions, production path

03

Source Code

Repository or packaged files, where applicable

04

Short README

How to run it, stack, assumptions, limitations

⌚ WHAT HAPPENS NEXT?

SUBMIT



MoE HR CONTACT



IN-PERSON PANEL · MoE DUBAI

Following submission, MoE Human Resources will contact you to arrange an in-person presentation and technical discussion at the Ministry of Education in Dubai. During the panel:

PRESENT

DEMONSTRATE

EXPLAIN

DEFEND

DISCUSS